

RESEARCH INTEREST

My research focuses on advancing optimization techniques for multi-objective deep neural networks, improving not only their helpfulness, but also their robustness and safety. I am particularly interested in exploring their effectiveness in large-scale applications, including recommender systems and large language models.

EDUCATION

- PhD** **Singapore University of Technology and Design (SUTD)** **09/2024 – Present**
- Academic advisors: [Prof. Wei Lu](#) and [Asst. Prof. Wenxuan Zhang](#)
 - Major: Computer Science.
- Bachelor** **Hanoi University of Science and Technology (HUST)** **08/2018 – 09/2022**
- Academic advisor: [Dr. Thang N. Tran](#)
 - Major: Mathematics and Informatics. CPA: 3.11/4.00 (Rank 18 out of 94 in my department).

PUBLICATIONS

- [1] Minh-Duc Nguyen, Phuong Mai Dinh, Quang-Huy Nguyen, **Long P. Hoang**, Dung D. Le, [Improving Pareto Set Learning for Expensive Multi-objective Optimization via Stein Variational Hypernetworks](#), *In Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
- [2] Tuan A. Tran[†], **Long P. Hoang**[†], Dung D. Le, Thang N. Tran, [A Framework for Controllable Pareto Front Learning with Completed Scalarization Functions and its Applications](#), *Neural Networks*, 2024.
- [3] **Long P. Hoang**, Dung D. Le, Tuan A. Tran, Thang N. Tran, [Improving Pareto Front Learning via Multi-Sample Hypernetworks](#), *In Proceedings of the AAAI Conference on Artificial Intelligence*, 2023 (Oral).
- [4] Anh T. Ho, Tuan A. Tran, **Long P. Hoang**, Ha H. Le, Thang N. Tran, [Multi Deep Learning Model for Building Footprint Extraction from High-Resolution Remote Sensing Image](#), *In Intelligent Systems and Networks*, 2022.

EXPERIENCE

- PhD Student** **StatNLP Reseach Group, SUTD**
- Advisor: [Prof. Wei Lu](#) and [Asst. Prof. Wenxuan Zhang](#)
- Personalized SFT Dataset for your Language Model* **04/2025 – Present**
- Developing a collaborative filtering framework to select a small, challenging, and diverse dataset personalized for a given language model.
 - Empirically validated that fine-tuning with 1,000 high-quality examples selected via our method yields better performance than using the entire 220,000-example dataset.
- Chain-of-Thought Behavior* **09/2024 – 02/2025**
- Studied token-level representations with and without Chain-of-Thought (CoT) prompting.
 - Identified a representational collapse phenomenon under CoT that enhances model confidence, emphasizes salient information, and mitigates gender bias.

[†]Co-First Author

Research Assistant

College of Engineering and Computer Science, VinUniversity

Advisor: [Asst. Prof. Dung D. Le](#)

Controllable Multi-Objective Recommender System

10/2022 – 10/2023

- Developed a new framework for Multi-Objective Recommendation which considers a variety of criteria, including fairness, robustness, novelty in special scenarios such as cold start, adversarial attack,...

Expensive Multi-Objective Optimization

03/2023 – 10/2023

- Approximated the entire trade-off curve of black-box objects using Pareto Front Learning with Hypernetworks computed by Gaussian Processes (accepted to AAAI 2025).

Profiling the Pareto Front in Multi-Task Learning

02/2022 – 10/2022

- Proposed a novel method named Multi-Sample Hypernetwork to approximate the entire trade-off curve of conflicting objectives (accepted to AAAI 2023).

Undergraduate Research Assistant

School of Applied Mathematics and Informatics, HUST

Advisor: [Dr. Thang N. Tran](#)

Multi-Objective Optimization with Completed Scalarizations

02/2022 – 02/2023

- Proposed and proved the convergence of a new framework for Pareto Multi-Task Learning with Scalarization Functions in the pseudo-convex and quasiconvex assumptions (accepted to the journal Neural Networks).

Building Footprint Extraction from Remote Sensing Images

07/2021 – 10/2022

- Developed a two-stage framework, which combines U2-Net and Mask-CNN, to increase 1.8-2.5% mAP, mAR for Building Footprint Extraction, especially effective in populated areas (accepted to ICISN 2022).

ADWARDS & CERTIFICATES

- AISG PhD Fellowship, granted by AI Singapore, Sep 2024 – Sep 2028.
- 3rd Prize (Grooo International company's sponsorship) in Scientific Research Student Conference at School of Mathematics and Informatics, Hanoi University of Science and Technology, Jul 2022.
- Certificate of Completion of Developer Circles Vietnam Innovation Challenge in Data Science, sponsored by Facebook, Nov 2020.

REFERENCES

1. [Prof. Wei Lu](#) (Ph.D), Information Systems Technology and Design, Singapore University of Technology and Design
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2. [Asst. Prof. Wenxuan Zhang](#) (Ph.D), Information Systems Technology and Design, Singapore University of Technology and Design
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3. [Asst. Prof. Dung D. Le](#) (Ph.D), College of Engineering and Computer Science, VinUniversity
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4. [Dr. Thang N. Tran](#) (Ph.D), School of Applied Mathematics and Informatics, Hanoi University of Science and Technology
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